

SECTION ONE: SCHEME OF ASSESSMENT

Candidates are required to enter for Papers 1, 2, and 3.

Paper	Type of Paper	Duration	Marks	Weighting
1	Multiple Choice	1 h	40	30 %
2	Structured and Free-Response Questions	1 h 45 min	80	50 %
3	Practical	1 h 50 min	40	20 %

Practical Paper 3 (1 h 50 min, 40 marks)

Scientific subjects are, by their nature, experimental. It is therefore important that an assessment of a candidate's knowledge and understanding of Science should include a component relating to practical work and experimental skills.

This assessment is provided in Paper 3 as a formal practical test and is outlined in the Scheme of Assessment.

This paper is designed to assess a candidate's competence in those practical skills which can realistically be assessed within the context of a formal test of limited duration.

This paper will comprise two to three compulsory practical questions, based mainly on Assessment Objective C.

One or more of the questions may incorporate assessment of Planning (P) and require candidates to apply and integrate knowledge and understanding from different sections of the syllabus. The assessment of PDO and ACE may include questions on data analysis which do not require practical equipment and apparatus.

Candidates are not allowed to refer to notebooks, textbooks or any other information during the assessment.

ASSESSMENT OBJECTIVES FOR PAPER 3	
Candidates should be able to: 1. follow a sequence of instructions; 2. use techniques, apparatus and materials; 3. make and record observations, measurements and estimates; 4. interpret and evaluate observations and experimental results; 5. plan investigations, select techniques, apparatus and materials; 6. evaluate methods and suggest possible improvements.	Paper 3 will assess appropriate aspects of objectives C1 to C6 in the following skill areas • Planning (P) • Manipulation, measurement and observation (MMO) • Presentation of data and observations (PDO) • Analysis, conclusions and evaluation (ACE) The assessment of Planning (P) will have a weighting of 15%. The assessment of skill areas MMO, PDO and ACE will have a weighting of 85%.

Candidates will be assessed in the following skill areas:

(a) Planning (P)

Candidates should be able to:

- identify key variables for a given question/problem
- outline an experimental procedure to investigate the question/problem
- describe how the data should be used in order to reach a conclusion
- identify the risks of the experiment and state precautions that should be taken to keep risks to a minimum

(b) Manipulation, measurement and observation (MMO)

Candidates should be able to:

- set up apparatus correctly by following written instructions or diagrams
- use common laboratory apparatus and techniques to collect data and make observations
- describe and explain how apparatus and techniques are used correctly
- make and record accurate observations with good details and measurements to an appropriate degree of precision
- make appropriate decisions about measurements or observations

(c) Presentation of data and observations (PDO)

Candidates should be able to:

- present all information in an appropriate form
- manipulate measurements effectively for analysis
- present all quantitative data to an appropriate number of decimal places/significant figures

(d) Analysis, conclusions and evaluation (ACE)

Candidates should be able to:

- analyse and interpret data or observations appropriately in relation to the task
- draw conclusion(s) from the interpretation of experimental data or observations and underlying principles
- make predictions based on their data and conclusions
- identify significant sources of errors and explain how they affect the results
- state and explain how significant errors may be overcome or reduced, as appropriate, including how
- experimental procedures may be improved.

One, or more, of the questions may incorporate some assessment of skill area P, set in the context of the syllabus content, requiring candidates to apply and integrate knowledge and understanding from different sections of the syllabus. It may also require the treatment of given experimental data in drawing relevant conclusion and analysis of proposed plan.

The assessment of skill areas MMO, PDO and ACE will be set mainly in the context of the syllabus content.

The assessment of PDO and ACE may also include questions on data-analysis which do not require practical equipment and apparatus.

The material used in experiments will be closely related to the subject matter of the syllabus but will not necessarily be limited to the particular types mentioned in the curriculum content.

SECTION TWO: PRECISION, ACCURACY, AND RELIABILITY

Precision

- 'Spread' of results. Precise results have very little deviation from the mean result
- The precision of a measurement depends on the ability of the measuring instrument to give you the same reading every time it measures the same thing.
- This does not have to be the 'true' value.
- Can be improved by using apparatus that has smaller scale divisions
- Precision can be improved by using apparatus that has smaller scale divisions.

Accuracy

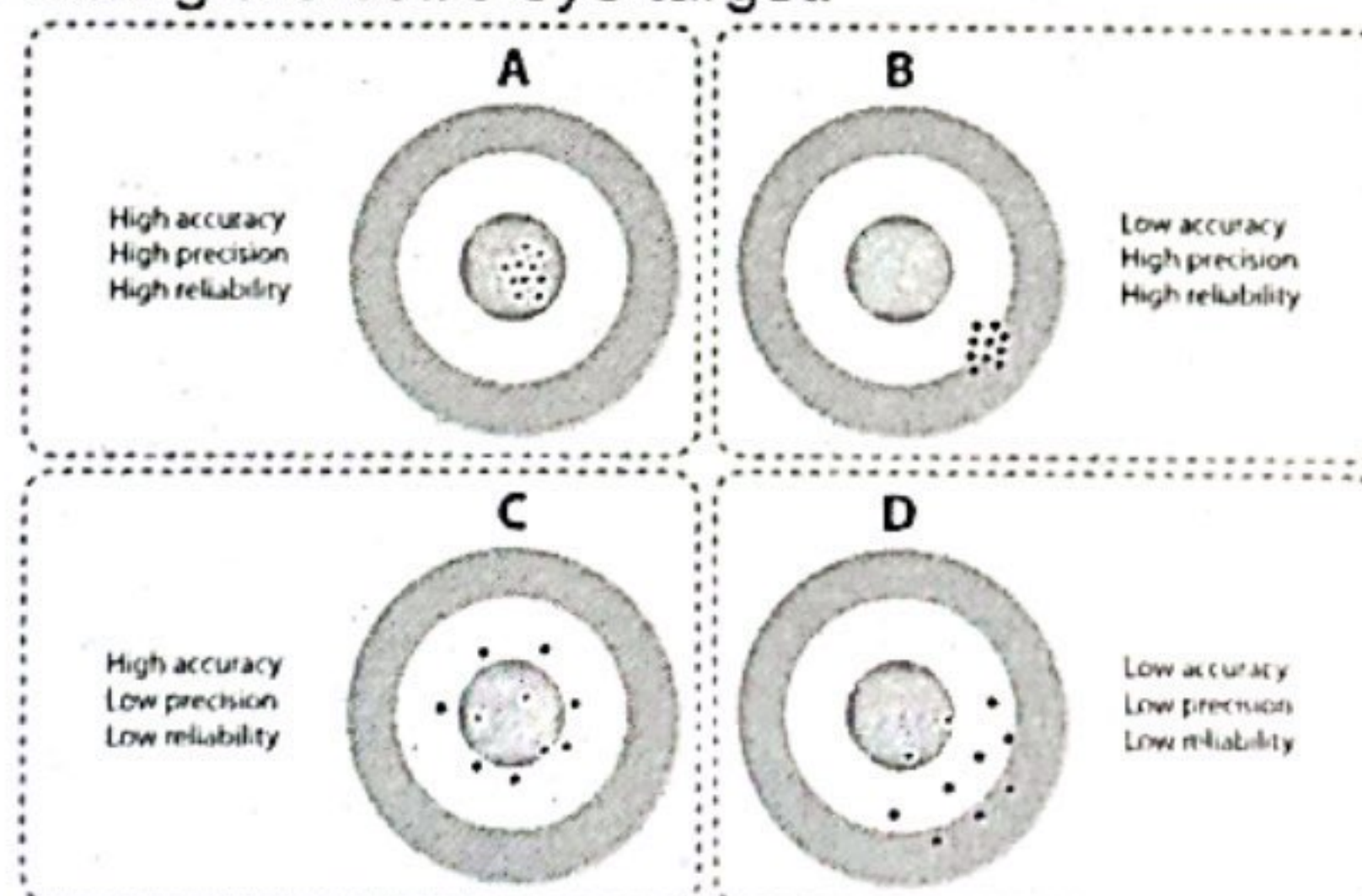
- Refers to how close the data or observation is to the actual 'true' value.
- Accuracy can be increased by eliminating or minimising errors.

Reliability

- Refers to the likelihood of getting the same results if you did the investigation again and being sure that the results are not due to chance.
- Also referred to as 'repeatability'.
- If the same result / readings are obtained when a measurement / entire experiment is repeated, it is considered reliable.
- Reliability of a single measurement can be improved by the following methods to limit the effect of random error:
 - Fixing controlled variables
 - Choosing more precise apparatus
- Reliability of the entire experiment can be improved by
 - Improving the reliability of single measurements
 - Increasing the number of repeats of each measurement and taking the average measurement

Visualisation of accuracy and reliability

As far as possible, results of an experiment should be precise, accurate, and reliable i.e. closely clustered and hitting the bull's eye target.

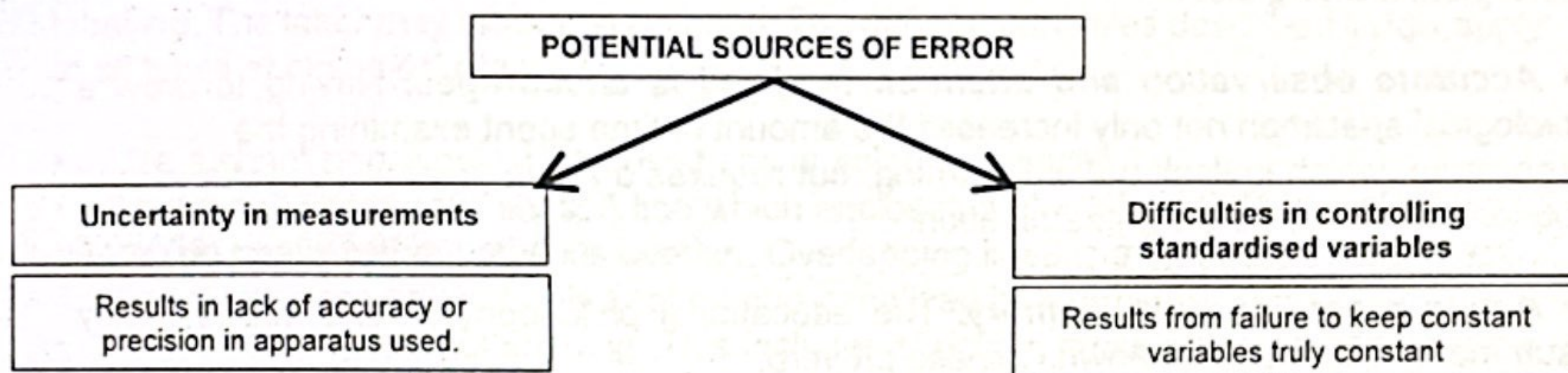


PRECISION OF INSTRUMENTS

No	Apparatus	Smallest Division	Uncertainty	Examples of recording
1	Burette	0.1 cm ³	0.05 cm ³	25.0 cm ³ 25.00 cm ³ , 25.05 cm ³
2	Electronic balance	0.1g	0.1 g	120.0 g, 121.1 g
		0.01 g	0.01 g	121.00 g, 121.10 g
3	Ruler	0.1 cm	0.1 cm	12.0 cm, 12.1 cm
4	Measuring cylinder (100 cm ³)	1 cm ³	0.5 cm ³	18.0 cm ³ , 18.5 cm ³
5	Stopwatch (analogue)	0.1 s	0.1 s	36.0 s, 36.1 s
6	Stopwatch (digital)	0.1 s	0.1 s	28.1 s
		0.01 s	0.01 s	28.00 s, 28.11 s
7	Thermometer (-10 °C to 110 °C)	1 °C	0.5 °C	23.0 °C, 23.5 °C

IDENTIFYING SOURCES OF ERROR AND SUGGESTING IMPROVEMENTS

- You are **NOT** being asked about the mistakes that you may have made during the experiment (e.g. not reading the thermometer correctly, taking measurements at the wrong time, parallax error) – these are human errors which can be avoided.
- Sources of error are *unavoidable limitations* of your apparatus, measuring instruments, experimental technique or experimental design that prevent your results from being totally reliable.



Possible improvements			
- Use more precise instruments - E.g. For volumes, use a graduated pipette instead of a syringe	- Using techniques or apparatus that are <i>better able</i> to keep standardised variables constant - E.g. using a thermostatically-controlled water bath instead of a beaker of water with Bunsen burner	- Controlling important variables which were not controlled in the original experiment. - State how it should be controlled.	Repeat the experiment at least 3 times so that several readings of the dependent variable can be obtained for each value of the independent variable. - Calculate the mean of the readings to obtain a more reliable result.

SECTION THREE: BIOLOGICAL DRAWINGS

Why bother?

The ability to draw, label and annotate biological specimens is an important and useful biological skill. These days students may well challenge the need for making biological drawings, particularly given the ease of using digital photography for record-keeping. So how can it be justified? The following points help to provide a rationale for developing biological drawing skills:

- ***Accurate observation and attention to detail is encouraged.*** Having to draw a biological specimen not only increases the amount of time spent examining the specimen, which in itself will aid learning, but requires a much greater level of accurate observation than a casual examination.
- ***Active recording aids memory.*** The educational philosophy behind this is neatly summarised in the well-known Chinese proverb:

*I hear and I forget
I see and I remember
I do and I understand*

Confucius

- ***The drawing provides a permanent record of what has been observed.*** There is a historic tradition within biology of providing accurate records of specimens so that the images could be used for future reference purposes.

Today's taxonomists are often indebted to the illustrators of the 17th and 18th centuries, particularly where the 'type' (reference) specimen may only exist as an illustration.

Even today, when digital photography can be used to store images, artists are still often commissioned to record biological specimens of interest by drawing or painting. This is particularly true for flowering plants. This is partly because all the features of interest can be combined in one or several scientifically accurate, but aesthetic, images with great clarity (see Figure 1).

You do not need to be an artist to be able to produce good biological drawings. It is a test of your observational skills.

What equipment is needed?

- Sharp pencil - 2B mechanical pencil is generally preferred.
- Pencil sharpener
- Eraser
- Ruler - For label lines.

General Principles

When assessing biological drawing, marks are awarded for both quality of drawing and labelling. The latter may include annotation. The general principles described below apply to all types of biological drawing:

- Use a sharp pencil only. Don't use pens or coloured pencils.
- Use clear, continuous lines. A line which encloses a shape, such as a circle, should join up neatly without obvious overlap. Overlapping lines is a common error in hastily drawn sketches and is easily spotted and penalized by examiners.
- Don't use any form of shading. This includes stippling, cross-hatching and shading. Students find this is a hard instruction to follow, and it is sometimes difficult to justify. Although shading may help to make the drawing look more realistic and/or to discriminate between areas of the specimen, it does not represent a permanent structural feature. Artistic impression is certainly not what is required.
- Accuracy is paramount. It shows good observation. Remember that observation is assisted by understanding, so a good knowledge of theory goes alongside good drawing. Pay particular attention to the outlines of structures and to the relative proportions of different parts of the specimen. Don't draw what you think you should see, for example text book style drawings. Draw what you observe.
- Guidelines can help. Faint sketching of the main areas of the specimen which can later be erased may help. Some students find a simple grid helps them.
- Make the drawing large enough. If the specimen is a relatively large structure such as a plant or a section of an organ, it should normally occupy more than half the available space on the page. In microscopy, individual cells drawn at high power should be about one to several centimetres in diameter.
- Correct mistakes. If you make a mistake, use a good quality eraser to rub out the lines completely.

Labelling

- Use a sharp pencil.
- Label lines should start exactly at the structure being labelled; don't use arrowheads
- Arrange label lines neatly and make sure they don't cross over each other. It is visually attractive, though not essential, if the length of the label lines is adjusted so that the actual labels are right or left justified, i.e. line up vertically above each other on either side of the drawing.
- Labels should be written horizontally, as in a textbook, not written at the same angle as the label line.

Pay attention to the following:

- Accuracy: drawing should resemble actual specimen in terms of size, shape, and proportion.
- Relevance: features drawn are related to aim of the task, e.g., in a task that requires students to draw parts of a wind pollinated flower, relevant features should include the different structural features of the stamen and stigma.

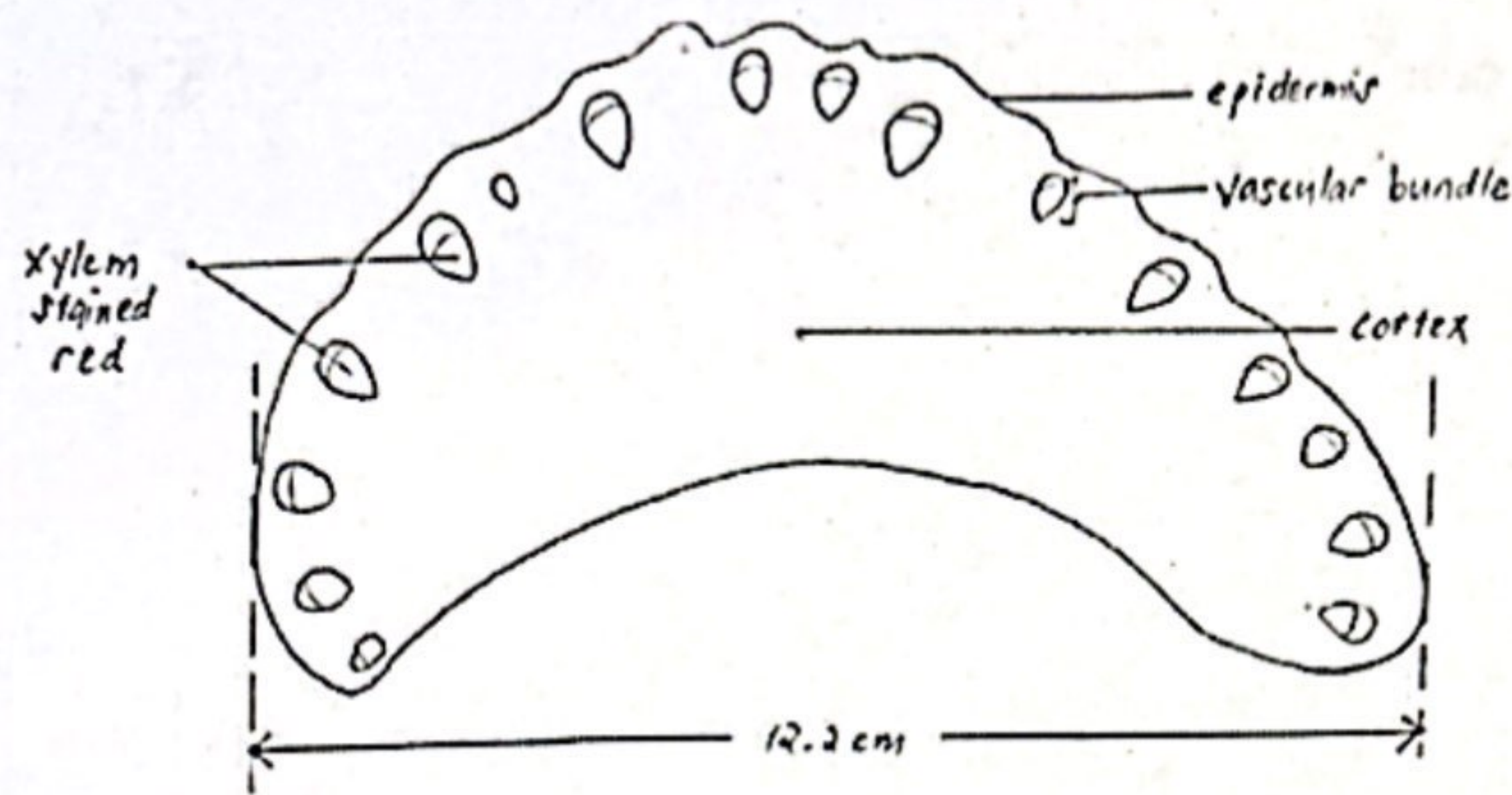
Tips on doing biological drawings:

- Make faint outlines of the specimen using a light pencil
- Use a ruler to check the proportions of the main outlines and mark proportions on your drawing
- Draw firm continuous lines around the main outlines and erase the faint construction lines drawn earlier
- Complete the drawing with finer details / anatomy
- Include labels if the question asks for it

MARKING CRITERIA for BIOLOGICAL DRAWINGS (SL²AP MED)

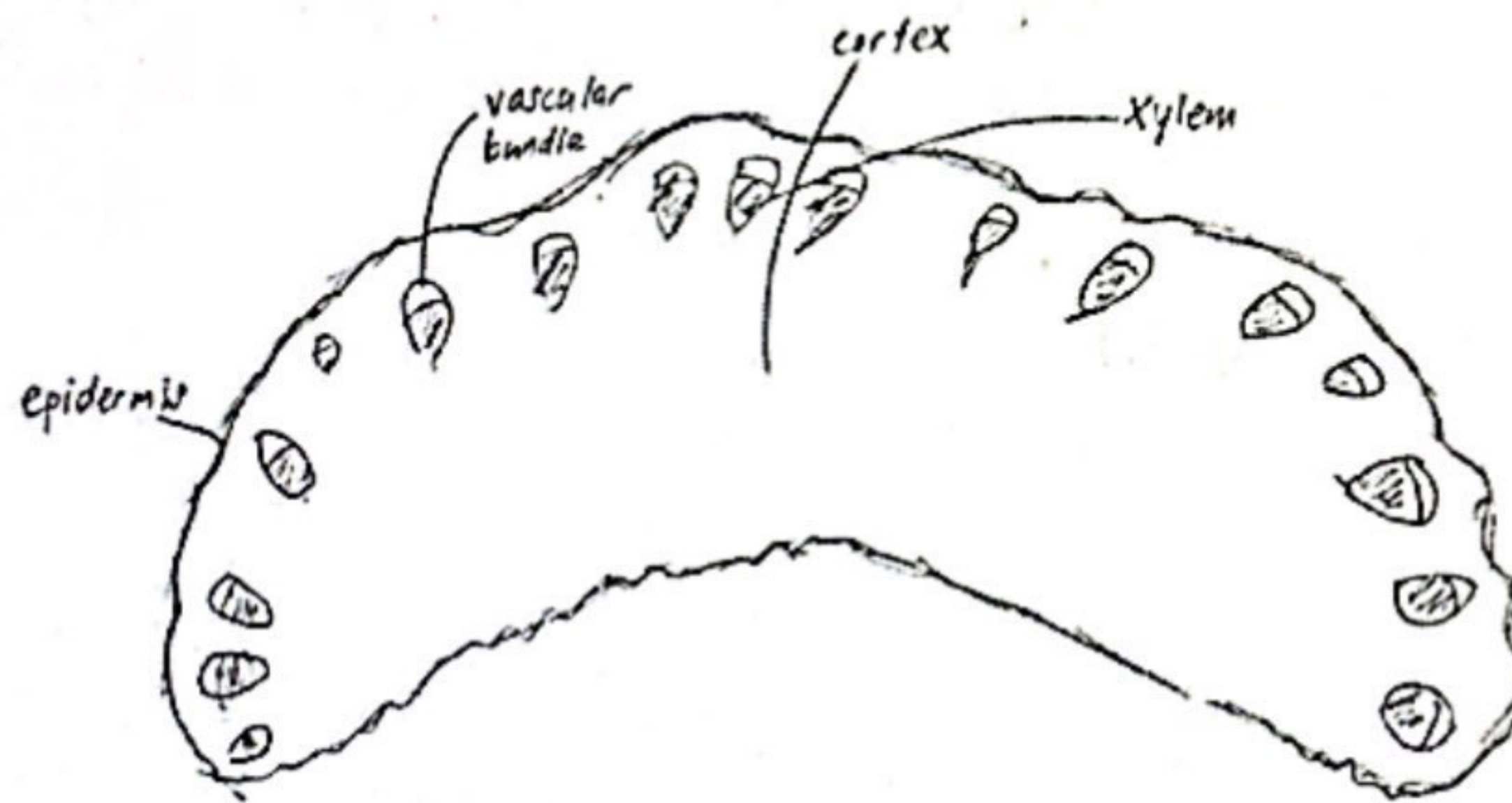
- ✓ Size: large drawing i.e. at least half of the space provided
- ✓ Line: clean, clear and sharp outline i.e. no fuzzy or broken line
- ✓ Labels: straight labelling lines i.e. draw using rule; and do not criss-cross one another
- ✓ Anatomy: details
- ✓ Proportionality
- ✓ Magnification
- ✓ Expression shown for calculation of magnification
- ✓ Decimal Places: up to 2 decimal places, not more than 0.2 rounding up/down

Example of acceptable biological drawing:



Transverse section of a celery stalk stained with red dye

Example of unacceptable biological drawing:



Unacceptable features:

- Label lines not straight
- Label lines cross each other
- Use of broken lines
- Shading

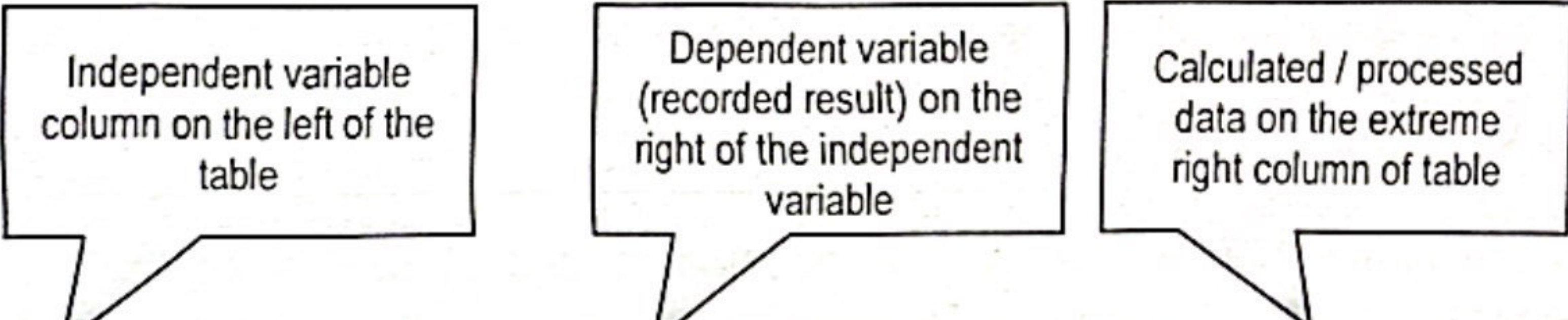
SECTION FIVE: DATA REPRESENTATION IN TABLES

The layout and contents of a table, whether it is for recording numerical data or observations, should be decided before the experiment is performed. 'Making it up as you go along' often results in tables that are difficult to follow and do not make the best use of space. Hence, it is crucial that students read and understand the entire experimental procedure before collecting any form of data.

The following are characteristics of a good table:

- 1) The independent variable should be in the leftmost column of the table, with following subsequent columns showing the dependent variables ;
- 2) The heading of each column must include both the quantity being measured and the units in which the measurement is made. The unit should be written in solidus (backslash) notation, e.g., the length of string is written as length / cm; the time taken for reaction as time / s ;
- 3) The body of the table must not contain units ;
- 4) Readings made directly from the measuring instruments should be given to the number of decimal places that is appropriate for the measuring instrument used.

Example of a good table:



test-tube	percentage concentration of chloroplast suspension	time taken for the purple-pink indicator to turn colourless / s	rate of photosynthesis / s ⁻¹
A	50	10	100
B	40	18	56
C	30	37	27
D	20	74	14
E	15	220	5
F	10	515	2

PRACTICE QUESTION 1

A student investigated the effect of different concentrations of sugar solutions on some potato cylinders. He cut out several cylinders from a large potato, each the same diameter. He then cut them into 1 cm lengths. He then measured the mass of each piece.

He placed one piece of potato into each of the six beakers and then covered each piece with different solutions of sugar solutions. He used the same volume for each beaker.

The student left the potato pieces in the beakers for 30 minutes. Then he removed them from the beakers, blotted them dry with paper and measured their mass again. These were the results he wrote down.

Before: piece A = 5.2g, piece B = 5.2g, piece C = 4.9g, piece D = 5.0g
piece E = 5.1g, piece F = 5.2g

Solutions:

A, distilled water B, 0.1% sugar solution C, 0.2% sugar solution
D, 0.5% sugar solution E, 0.8% sugar solution, F, 1.0% solution

After: A = 5.5g B = 5.2g C = 4.9g D = 5.3g E = 5.0g
F = 5.0g

In the space below, draw a results table for the student's results. Include a column showing the change in mass. Decide if there are any anomalous results. If you think there are, draw a ring around them.

SECTION SIX: DATA REPRESENTATION IN GRAPHS

Presenting data in a graphical form helps us see relationships between variables more easily. The common types of graphs in practical include:

- (a) Line graphs
- (b) Bar charts
- (c) Histograms

Line Graphs

Line graphs are very common in practical work. A well-drawn line graph can help you identify relationships between the variables and see patterns.

- (a) Graphs should be drawn with a fine-tipped writing instrument e.g. mechanical pencil
- (b) Axes must be labelled clearly with quantity and unit
- (c) Data points should be indicated with a small cross
- (d) Scales for axes should be chosen to maximise the use of the graph paper in the vertical and horizontal directions. This allows a large graph to be plotted.
- (e) Use a sensible ratio for scale. Only use multiples of 2, 5 or 10 to indicate unit divisions on both axes

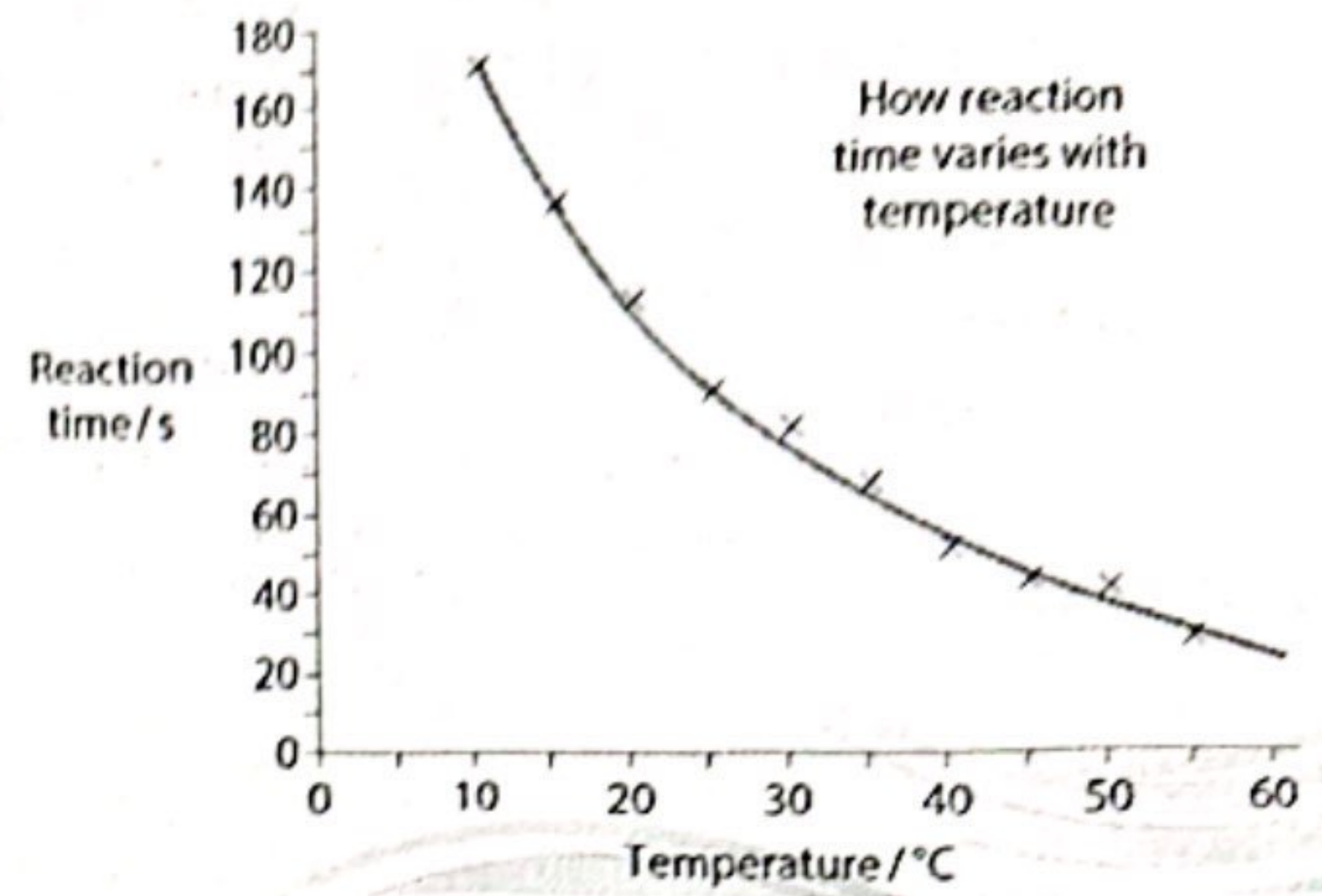
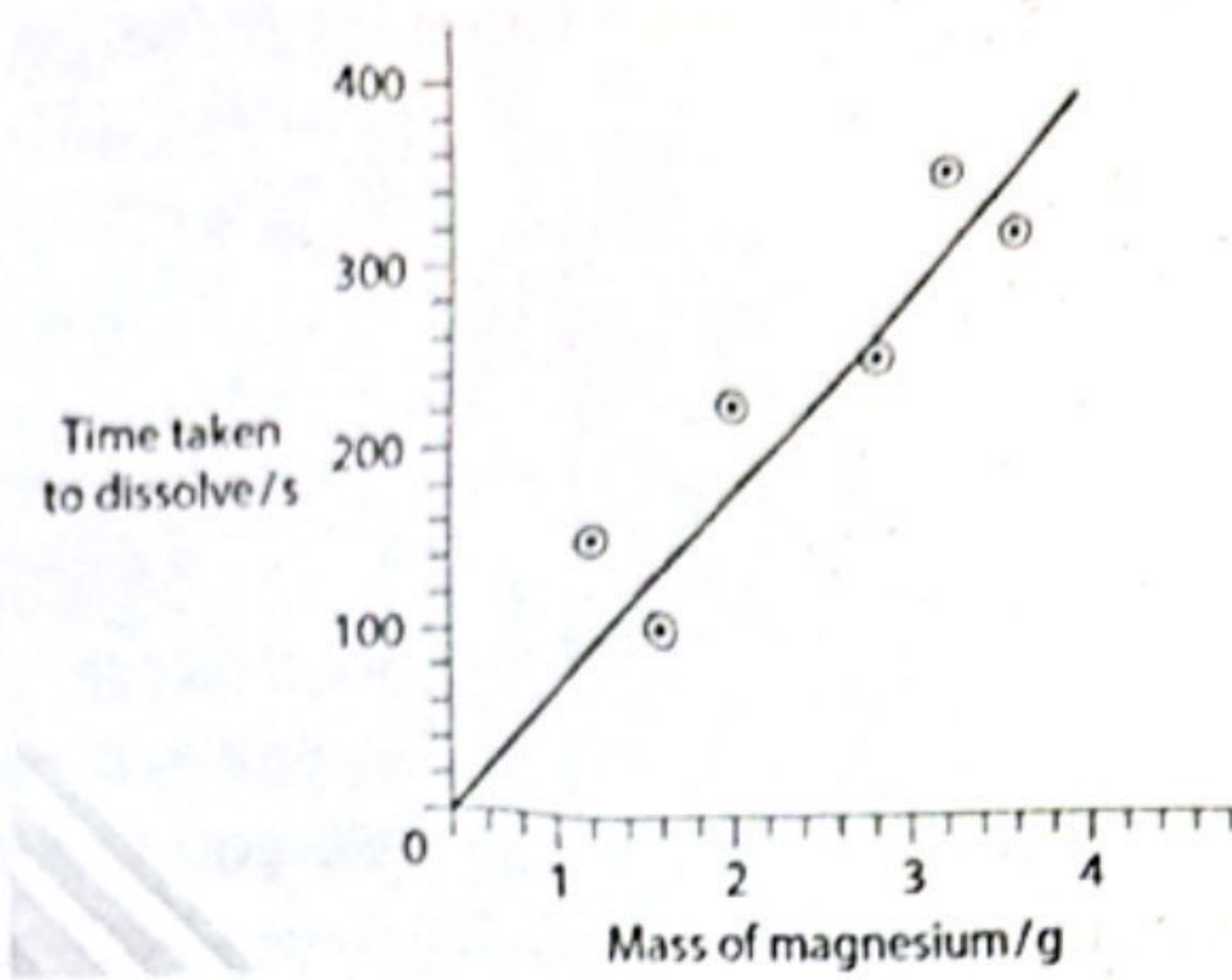
Drawing a line of best fit:

- (a) A line of best fit, which may be a straight line or curve, should be drawn when there is a possible relationship between the variables
- (b) It should have about equal number of points on either side of the line over its entire length
- (c) This line may or may not pass through any of the plotted points
- (d) The line drawn should be smooth and even to facilitate the accurate reading of coordinates along the line. Students are encouraged to use a transparent ruler when drawing the line of best fit so that they can see the other points through the ruler. If a curve is to be drawn, students are encouraged to use a flexi-curve or transparent curve ruler to help them with curve drawing.

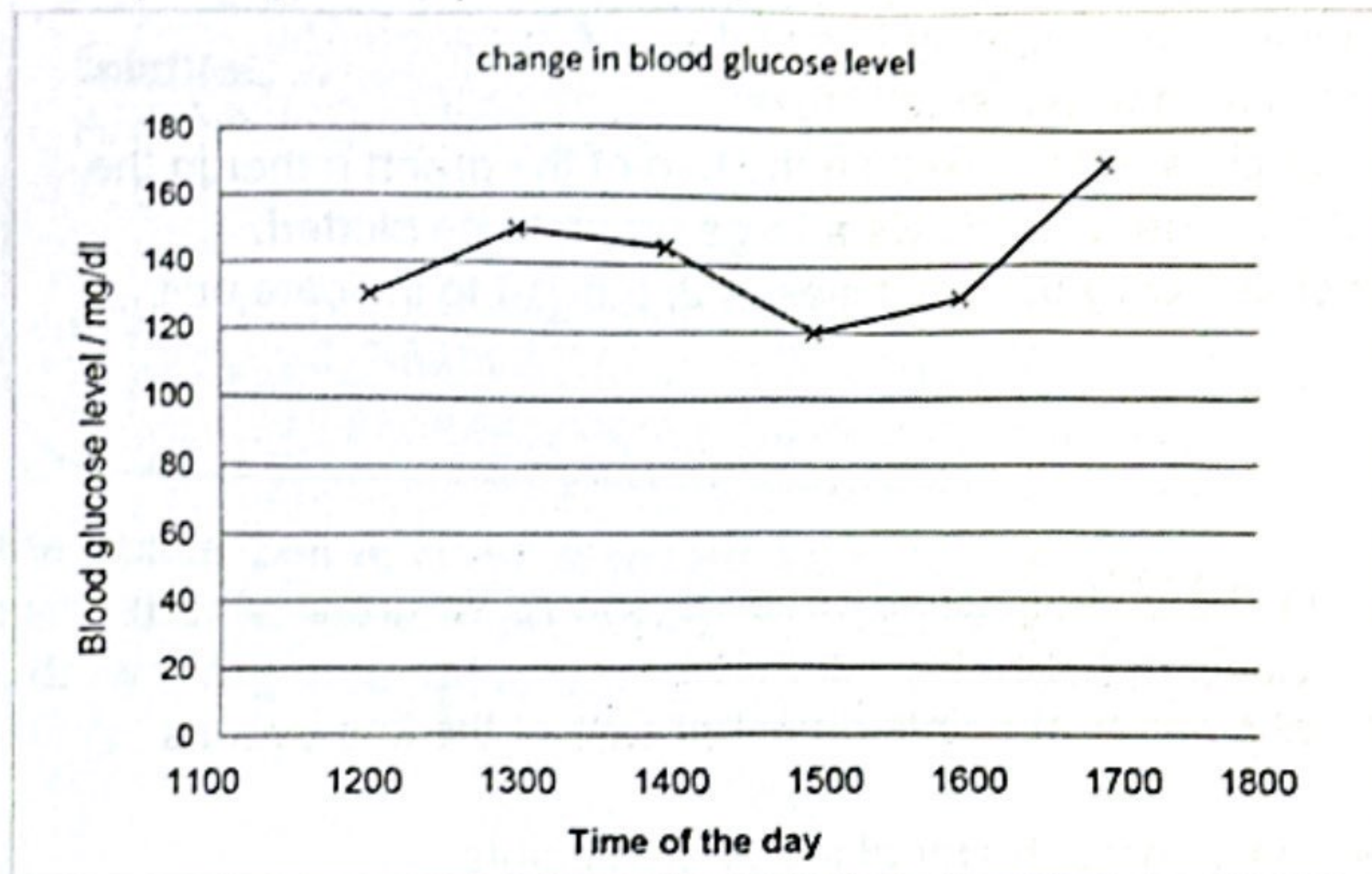
Drawing a line that joins the data from one point to another:

- (a) A point to point line should be drawn when there is no possible known relationship between the two variables, or there is no investigation to be done to determine the relationship between the variables.

Example of best fit lines:

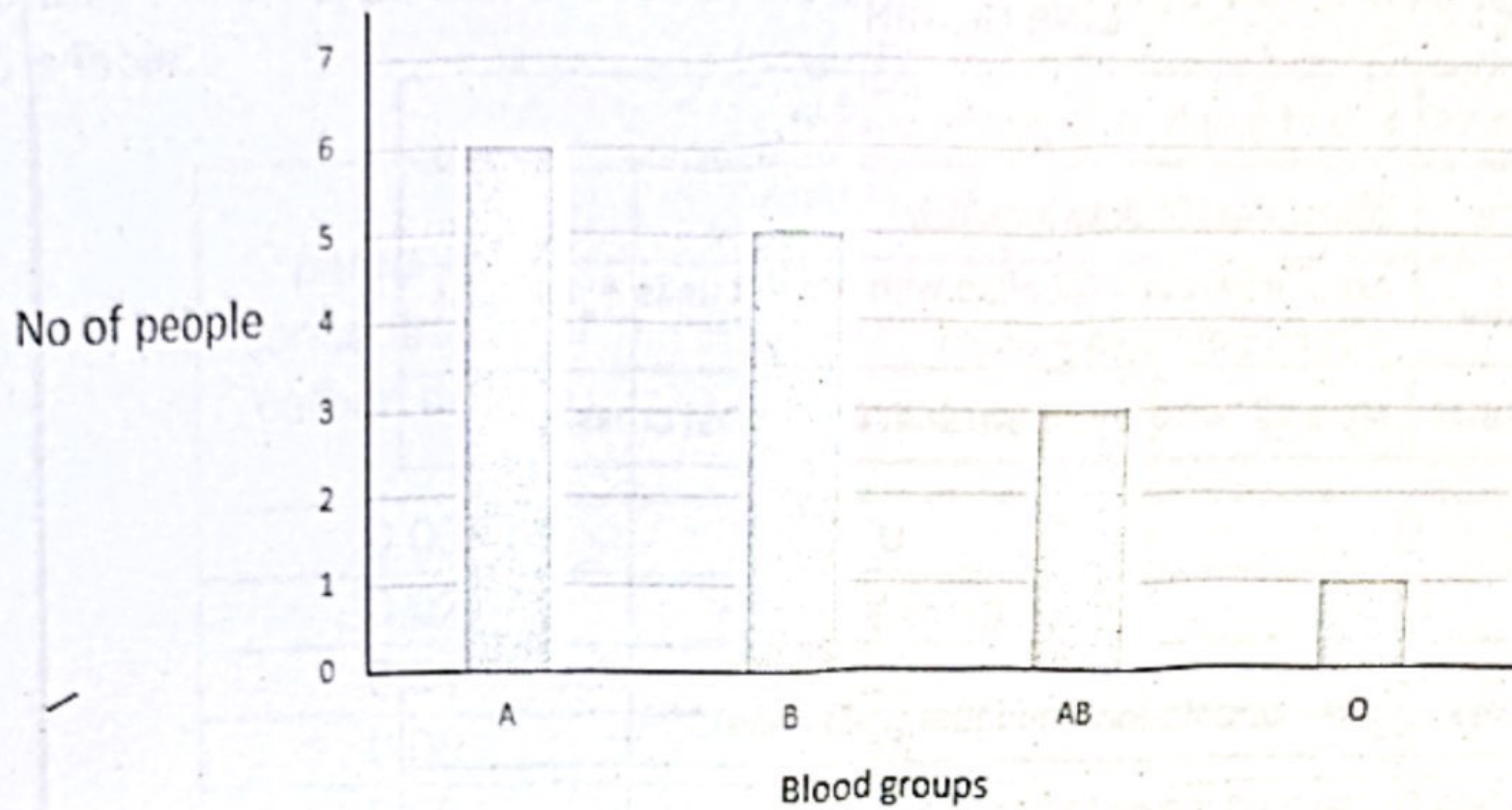


Example of point-to-point:



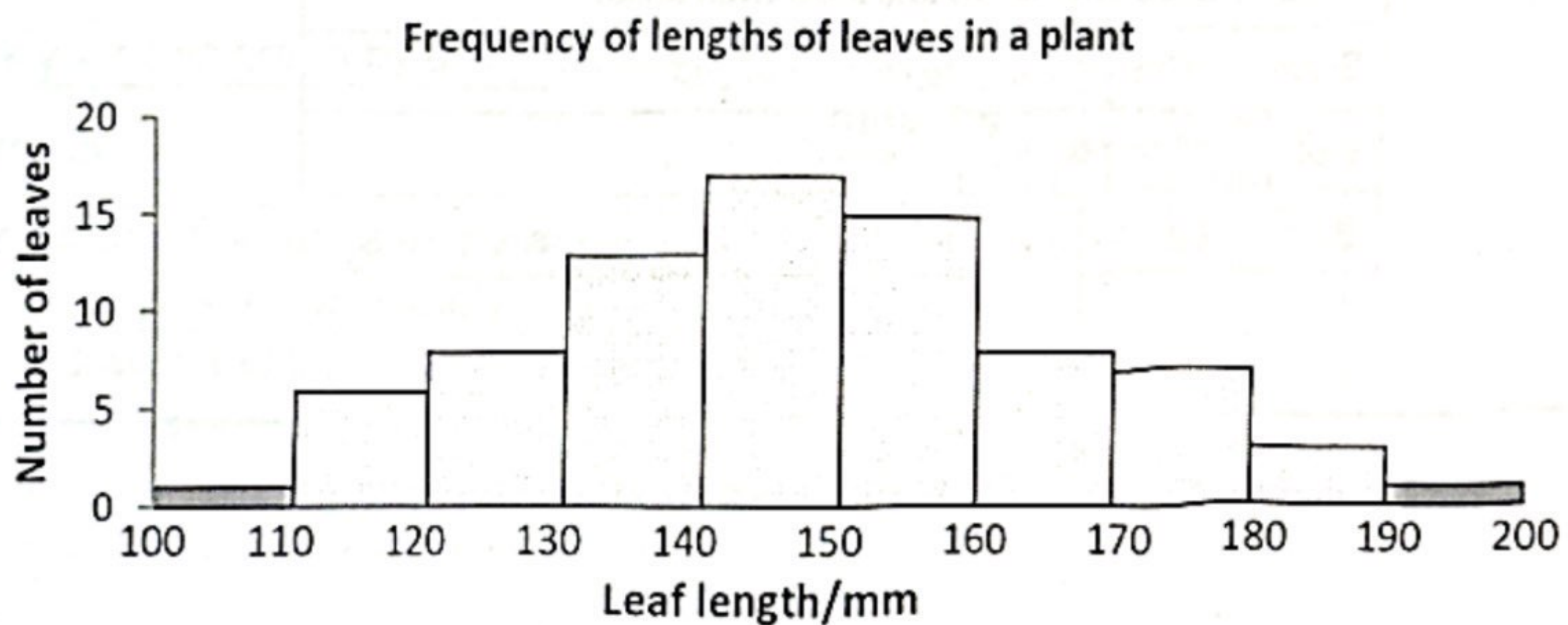
Bar charts

Bar charts are used when the independent or dependent variable is categorical.



Histograms

Continuous and numerical data can be represented in a histogram. This is different from a bar chart as each of these blocks touches the next and is ordered in either increasing or decreasing magnitude.



Marking Points for graphs

LINE GRAPH

Scale	size of graph as large as possible
Line	line of best fit drawn clearly
Axes	axes drawn and labelled with correct units + divisions with proper scale + origin
Plot	plots all correctly drawn with small neat crosses

BAR CHART

Axes	axes orientation and label (with units)
Scale	scale on y-axis to fill most of grid
Plot	all points plotted correctly
Bars	bars <u>not</u> touching and spaced equally on x-axis

HISTOGRAM

Axes	axes orientation and label (with units)
Scale	scale on y-axis to fill most of grid
Plot	all points plotted correctly
Bars	<u>bars touching</u> and spaced equally on x-axis

SECTION SEVEN: PLANNING

For the planning question, you are expected to be able to:

1. Identify key variables (independent, dependent, constant variables) for a given question
2. Outline an experimental procedure clearly to investigate the question / problem
3. Describe how the data should be used in order to reach a conclusion

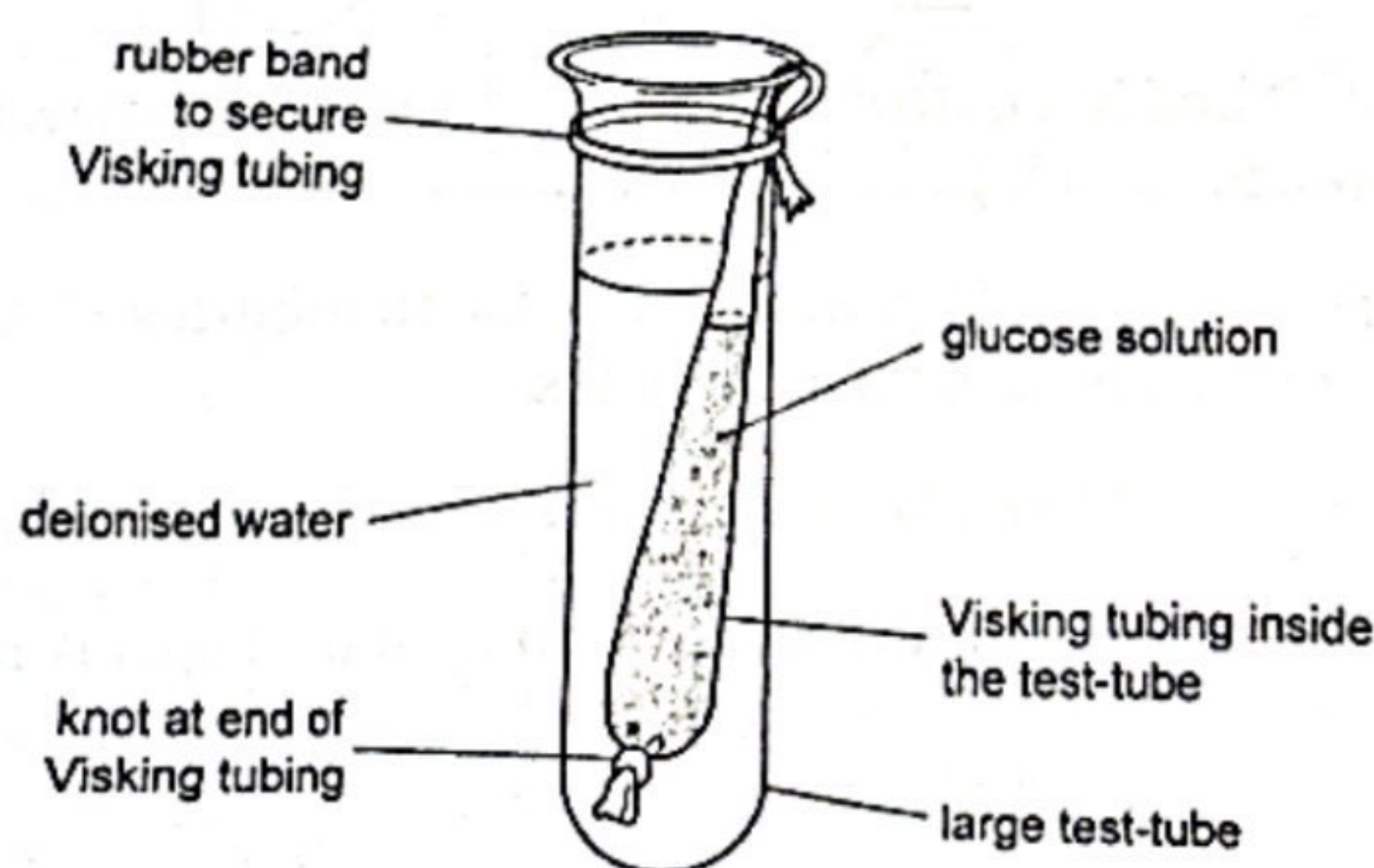
You are **not** to carry out the experiment you have planned. Instead, it would entail planning an investigation of a *different variable* from the earlier part of the question.

It would be wise to plan your answer by identifying the variables and apparatus you would be using. Refer to example below.

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In the earlier part of the question, candidates were required to carry out the Benedict's test to estimate the concentration of glucose in 'urine' samples to determine which test subject was diabetic. The concentration of glucose in the samples were estimated by recording the time taken for a **colour change to first occur** on the surface of the Benedict's solution.

- (g) Visking tubing is similar to the tubing used in kidney dialysis. Glucose diffuses through Visking tubing. A student set up the apparatus shown below to investigate the effect of temperature on the rate of diffusion of glucose. Plan an investigation **based on the apparatus** in the figure to determine the **effect of temperature** on the **rate of diffusion of glucose through Visking tubing**. [4]



You should make use of the blank spaces on the question booklet to identify the variables and the apparatus you would use in your answer. An example of planning:

Type of variable	Variable	Apparatus
Independent variable	Temperature	Using thermostatically-controlled water bath - 20 °C, 25 °C, 30 °C, 35 °C, 40 °C
Dependent variable	Record first colour change to first occur on surface of Benedict's solution	Stopwatch
Constant variables	Incubation duration	Stopwatch
	Volume of deionized water used for Benedict's test	Syringe
	Volume of glucose solution used	Syringe

You can start to craft your answer using the details you have planned above.

SAMPLE ANSWER

Step 1: Prepare 5 sets of the apparatus as shown in Fig. 1.2. Use a **syringe** to place **10 cm³ glucose solution of the same concentration** into the Visking tubing of each of the 5 set-ups (C.V.).

Step 2: Prepare 5 **thermostatically-controlled** water baths with the following temperatures: **20 °C, 25 °C, 30 °C, 35 °C, and 40 °C.** (I.V.)

Step 3: Immerse each set up into one of the water baths for **10 minutes** (C.V.). Start **stop-watch** to measure each set-up is only immersed for 10 minutes.

Step 4: Using a **syringe**, remove **2 cm³** of the deionised water (C.V.) from each large test-tube.

Step 5: Perform Benedict's test and start the **stopwatch** as the sample is placed into the water bath. Record the time taken for a colour change to first occur on the surface of the Benedict's solution using a stopwatch. (D.V.) Record results in a table.

Step 6: Repeat the experiment **3 more times** and **calculate the mean time taken** for a colour change to first occur for each temperature.

Step 7: The shortest time taken for the Benedict's solution to change colour shows the fastest rate of glucose diffusing through Visking tubing. OR Using the mean time taken for colour change to first occur and the graph on Fig. 1.1, deduce the concentration of glucose in the deionised water. Plot a graph of concentration of glucose VS temperature.